FTE4312 HW2 文杰 123090612

A

1. T
   - **Explanation:** In Uniswap V2, the standard 0.3% fee is deducted from the incoming token before the constant product formula ($x \times y = k$) is satisfied. This fee is then added to the pool's reserves. Because the reserves grow by the fee amount, the new product of the reserves (k) will be slightly larger than it was before the swap.
2. F
   - **Explanation:** Uniswap v2 requires liquidity to be deposited in the exact same proportion as the current pool reserves. If you attempt to supply tokens in a different ratio, the smart contract will only accept the tokens up to the correct ratio, and any excess of the over-supplied token will be left over (or the transaction will fail, depending on the router used).
3. T
   - **Explanation:** The $x \times y = k$ curve is asymptotic. As a trade depletes a significant portion of one token's reserve, the remaining tokens become exponentially more expensive. This causes the execution price to deviate significantly from the spot price, resulting in high slippage (price impact).
4. F
   - **Explanation:** The liquidation threshold is typically higher than the maximum borrow limit (often called the Loan-to-Value or LTV limit). This gap creates a safety buffer. For example, a protocol might let you borrow up to 75% of your collateral's value, but only liquidate you if your debt reaches 80%, giving you a window to manage your risk before being penalized.
5. T
   - **Explanation:** This is the standard formula used by major lending protocols like Aave and Compound. The utilization rate is a critical metric because it dictates the interest rate model; as utilization approaches 100%, borrow rates spike to incentivize repayment and attract new suppliers.
6. F
   - **Explanation:** A flash loan must be borrowed, utilized, and fully repaid within the exact same Ethereum transaction (which inherently means the exact same block). If the required funds plus fees are not returned to the pool by the end of the transaction execution, the entire transaction reverts as if it never happened.
7. T
   - **Explanation:** Arbitrageurs constantly monitor pools for discrepancies. If an AMM prices an asset lower than a centralized exchange, arbitrageurs will buy the asset from the AMM (increasing its price on the AMM due to the bonding curve) and sell it on the exchange for a risk-free profit. They will continue doing this until the prices reach equilibrium.
8. F
   - **Explanation:** A sandwich attack consists of two parts: *front-running* and *back-running*. The attacker places their front-running transaction *before* the victim's trade to drive the price up. The victim then buys at the inflated price. Finally, the attacker places their back-running transaction *after* the victim's trade to sell their assets at a profit.
9. T
   - **Explanation:** To ensure the system remains solvent, protocols need third parties to aggressively clear undercollateralized debt. They incentivize this by allowing liquidators to pay off the borrower's debt in exchange for receiving the borrower's collateral at a discount (e.g., a 10% bonus), yielding an immediate profit.
10. F
    - **Explanation:** The transition to PoS (The Merge) eliminated *miners*, but it did not eliminate the underlying mechanics of transaction ordering. The power to order, include, or exclude transactions shifted to *validators* (block proposers). MEV was simply rebranded to "Maximal Extractable Value," and it remains a massive part of the Ethereum ecosystem through infrastructure like MEV-Boost.

B

1. A
   - **Explanation:** In a constant-product AMM, the spot price is calculated as the ratio of the two reserves. Dividing the USDC reserve by the ETH reserve (100,000 / 50) gives a spot price of 2,000 USDC per 1 ETH.
2. B
   - **Explanation:** Impermanent loss represents the opportunity cost of providing liquidity compared to just holding the assets. As relative prices change, the AMM automatically rebalances by selling the outperforming asset and buying the underperforming one, resulting in a lower portfolio value than holding the original deposits.
3. A

- **Explanation:** The collateral factor (often referred to as the maximum Loan-to-Value or LTV ratio) is a protocol-defined parameter that sets the exact percentage of your collateral's total value that you are allowed to borrow.

4. B
   - **Explanation:** Utilization is calculated as the ratio of total borrowed funds to total supplied funds. Dividing 600,000 DAI by 1,000,000 DAI yields a 60% utilization rate.
5. C
   - **Explanation:** Using the provided formula: BorrowAPR = 2% + (20% * 0.60) = 2% + 12% = 14%.
6. B
   - **Explanation:** In Uniswap v2, the fee is deducted from the input amount before calculating the output. The formula applies the $(1 - f)$ multiplier to `amountIn`, which is derived algebraically from the constant product invariant $x \cdot y = k$ to solve for the exact output amount.
7. A
   - **Explanation:** A health factor below 1 means that the adjusted value of the borrower's collateral (factoring in the liquidation threshold) has fallen below the value of their borrowed debt, meaning the position is undercollateralized and safe to be liquidated by bots.
8. B
   - **Explanation:** The standard arbitrage strategy is to buy low and sell high. Buying the discounted ETH on Pool A and immediately selling it for a premium on Pool B secures a risk-free profit.
9. C
   - **Explanation:** Flash loans rely entirely on the atomic nature of blockchain smart contracts. If the repayment condition is not met at the very end of the execution block, the entire transaction fails and reverts, acting as if the borrow never occurred.
10. B
    - **Explanation:** When an asset trades below its known peg on a specific exchange, rational arbitrageurs will buy the underpriced asset (DAI at 0.98 USDC) and sell or redeem it elsewhere at its true value (1.00 USDC) to capture the spread.

C

C1

For a Uniswap v2-style constant-product AMM, use the fee-adjusted input:

$$\Delta x_{\text{eff}} = \Delta x(1 - f) = 5000(1 - 0.003) = 4985$$

The output formula is:

$$\Delta y = \frac{\Delta x_{\text{eff}} \cdot y}{x + \Delta x_{\text{eff}}}$$

Uniswap v2 pricing is based on the constant-product liquidity-pool rule, and v2 charges a flat 0.30% swap fee added to reserves. The course materials also describe Uniswap as a DEX swap application in the Ethereum/Smart Contract topic.

a) ETH received

$$\Delta ETH = \frac{4985 \times 25}{50000 + 4985} \approx 2.2665$$

b) New reserves

USDC reserve increases by the full input amount:

$$50000 + 5000 = 55000$$

ETH reserve decreases by the ETH output:

$$25 - 2.2665 = 22.7335$$

c) Spot price before and after

Before:

$$P_0 = \frac{50000}{25} = 2000$$

After:

$$P_1 = \frac{55000}{22.7335} \approx 2419.34$$

d) Minimum ETH output with 1% slippage tolerance

If expected output is 2.2665 ETH, then minimum acceptable output is:

$$2.2665 \times (1 - 0.01) \approx 2.2439$$

C2

Assumption Note for Grading:

To satisfy the problem's dual constraints—that the pool is **100 ETH** and **200,000 DAI**, and that Alice's deposit of **10 ETH** and **20,000 DAI** exactly equals a **10%** share—we must assume the stated pool size represents the total reserves *after* Alice's initial deposit.

a) Compute the new pool reserves (ETH and DAI)

In a standard Automated Market Maker (AMM) using the constant product formula, the reserves satisfy the equation $x \cdot y = k$, where x is the ETH reserve and y is the DAI reserve.

Initial pool reserves are given as **100 ETH** and **200,000 DAI**.

$$k = 100 \cdot 200,000 = 20,000,000$$

The spot price P is determined by the ratio of the reserves. After arbitrage, the new price is **2,500 DAI/ETH**.

$$\frac{y_{new}}{x_{new}} = 2500 \implies y_{new} = 2500 \cdot x_{new}$$

Substitute y_{new} into the constant product equation:

$$x_{new} \cdot (2500 \cdot x_{new}) = 20,000,000$$

$$\therefore x_{new} = \sqrt{8000} \approx 89.4427$$

$$\therefore y_{new} = 2500 \cdot 89.4427 \approx 223,606.80$$

Answer (a): The new pool reserves are approximately **89.44 ETH** and **223,606.80 DAI**.

b) How many ETH and DAI does Alice own via her LP shares?

Alice holds exactly **10%** of the pool's liquidity shares. Her claim on the pool's current assets is strictly proportional to her share.

- **Alice's ETH:** $89.4427 \cdot 0.10 = 8.94427$
- **Alice's DAI:** $223,606.80 \cdot 0.10 = 22,360.68$

Answer (b): Alice owns approximately **8.94 ETH** and **22,360.68 DAI**.

c) Compute Alice's LP position value vs. holding, and the Impermanent Loss

Evaluate Alice's current assets at the new market price of **2,500 DAI/ETH**.

$$V_{LP} = (8.94427 \cdot 2500) + 22,360.68 = 44,721.36 \text{ DAI}$$

Evaluate the value of Alice's initial deposit (**10 ETH** and **20,000 DAI**) if she had held them in her wallet rather than providing liquidity, evaluated at the new market price.

$$V_{HODL} = (10 \cdot 2500) + 20,000$$

$$V_{HODL} = 25,000 + 20,000 = 45,000 \text{ DAI}$$

Impermanent loss is the percentage difference between the LP value and the HODL value.

$$IL = \frac{V_{LP} - V_{HODL}}{V_{HODL}} = \frac{44,721.36 - 45,000}{45,000} \approx -0.00619$$

Answer (c): Alice's LP position value is **44,721.36 DAI**, compared to a HODL value of **45,000 DAI**. Her impermanent loss is approximately **-0.62%**.

C3

a) ETH received from Pool A

$k_A = 100 \times 200,000 = 20,000,000$
$x_{new} = \frac{20,000,000}{200,000+5,000} \approx 97.561$
Result: $\Delta x = 100 - 97.561 \approx \mathbf{2.439}$ ETH

b) USDC received from Pool B and Profit

$k_B = 100 \times 220,000 = 22,000,000$
$y_{new} = \frac{22,000,000}{100+2.439} \approx 214,761.905$
Result (USDC Received): $\Delta y = 220,000 - 214,761.905 \approx \mathbf{5,238.10}$ USDC
Result (Profit): $5,238.10 - 5,000 = \mathbf{238.10}$ USDC

c) Post-Arbitrage Spot Prices

Result (Pool A): $P_A = \frac{205{,}000}{97.561} \approx \mathbf{2,101.25}$ USDC/ETH
Result (Pool B): $P_B = \frac{214{,}761.905}{102.439} \approx \mathbf{2,096.49}$ USDC/ETH

C4

a) Health factor after ETH drops

Collateral value:

$$10 \times 1650 = 16500 \text{ DAI}$$

Adjusted collateral value:

$$16500 \times 0.8 = 13200 \text{ DAI}$$

Health factor:

$$\text{HF} = \frac{13200}{14000} = 0.942857 \approx \boxed{0.943}$$

Since

$$0.943 < 1$$

Alice is liquidatable.

b) Maximum debt without liquidation

Need:

$$\text{HF} \geq 1$$

So:

$$\text{Debt} \leq \text{Collateral Value} \times \text{Liquidation Threshold}$$

$$\text{Max Debt} = 16500 \times 0.8 = \boxed{13200 \text{ DAI}}$$

c) Debt that must be repaid to restore HF to 1

Current debt:

$$14000$$

Target debt for HF = 1:

$$13200$$

Required repayment:

$$14000 - 13200 = \boxed{800 \text{ DAI}}$$

So Alice must repay 800 DAI.

d) Liquidator repays 5,000 DAI; liquidation bonus = 5%

Liquidator receives collateral worth:

$$5000 \times (1 + 5\%) = 5000 \times 1.05 = 5250 \text{ DAI}$$

Convert this DAI value into ETH at 1,650 DAI/ETH:

$$\text{ETH seized} = \frac{5250}{1650} \approx 3.182$$

So the liquidator can seize about 3.182 ETH.

C5

a)

$$u = \frac{600{,}000}{1{,}000{,}000} = 0.6 = 60$$

$$\text{BorrowAPR} = 2$$

$$\text{SupplyAPR} = 14$$

So:

$$\boxed{u = 60\%, \quad \text{BorrowAPR} = 14\%, \quad \text{SupplyAPR} = 7.56\%}$$

b) Lender interest for 30 days:

$$50{,}000 \times 7.56 = 310.6849$$

c) Borrower interest for 30 days:

$$10{,}000 \times 14 = 115.0685$$

C6

a) Attacker front-runs with 5 ETH

After attacker inputs 5 ETH:

$$x' = 100 + 5 = 105$$
$$y' = \frac{30000000}{105} = 285714.2857$$

So attacker receives:

$$300000 - 285714.2857 = 14285.7143$$

b) Victim swaps 20 ETH after attacker

Victim adds 20 ETH:

$$x'' = 105 + 20 = 125$$
$$y'' = \frac{30000000}{125} = 240000$$

Victim receives:

$$285714.2857 - 240000 = 45714.2857$$

c) Victim output with no attacker

If victim swaps directly into the initial pool:

$$x' = 100 + 20 = 120$$
$$y' = \frac{30000000}{120} = 250000$$

Victim receives:

$$300000 - 250000 = 50000$$

So the attacker's presence reduces victim output by:

$$50000 - 45714.2857 = 4285.7143$$

d) Attacker back-runs: swaps all USDT back to ETH

Before back-run, pool is:

$$(125 \text{ ETH}, \ 240000 \text{ USDT})$$

Attacker inputs:

$$14285.7143 \text{ USDT}$$

New USDT reserve:

$$240000 + 14285.7143 = 254285.7143$$

New ETH reserve:

$$x''' = \frac{30000000}{254285.7143} = 117.9775$$

So attacker receives:

$$125 - 117.9775 = 7.0225$$

Attacker started with 5 ETH, so profit is:

$$7.0225 - 5 = 2.0225$$

C7

Use

$$\text{HF} = \frac{\text{collateral value} \times \text{liquidation threshold}}{\text{debt}}$$

a) ETH price where Bob becomes liquidatable

Bob has 2 ETH collateral and 2,500 DAI debt.

Let ETH price be P DAI/ETH.

$$\text{collateral value} = 2P$$

Liquidation threshold = $80\% = 0.8$.

Set $\text{HF} = 1$:

$$1 = \frac{2P \times 0.8}{2500}$$

$$\boxed{P = 1562.5 \text{ DAI/ETH}}$$

So Bob becomes liquidatable when ETH falls to 1,562.5 DAI/ETH or below.

b) Maximum initial borrow with 75% collateral factor

Initial ETH price = 2,000 DAI/ETH.

Collateral value:

$$2 \times 2000 = 4000 \text{ DAI}$$

Collateral factor = $75\% = 0.75$.

$$\text{max borrow} = 4000 \times 0.75 = 3000$$

c) HF at ETH = 1,400 and repayment needed

Collateral value:

$$2 \times 1400 = 2800 \text{ DAI}$$

Adjusted collateral value under liquidation threshold:

$$2800 \times 0.8 = 2240$$

Debt = 2,500 DAI.

$$\text{HF} = \frac{2240}{2500} = 0.896$$

Since $\text{HF} < 1$, Bob is liquidatable.

To bring HF back to 1 while keeping collateral fixed, the new debt must equal the adjusted collateral value:

$$1 = \frac{2240}{\text{new debt}}$$

$$\text{new debt} = 2240$$

Current debt is 2,500 DAI, so required repayment:

$$2500 - 2240 = 260$$